

BIOL-1 Practice Test 05

Comprehensive Final Review (Modules 01–15)

College of the Redwoods — BIOL-1: General Biology (Pelican Bay, Fall 2026)

Instructions: This practice test samples **big ideas** from every module. Work through Parts A–C. For Part C, answer in complete sentences.

Part A: Multiple Choice (75 questions)

Choose the best answer for each question.

Module 01: Study of Life

1. Which property is shared by every living organism in this course's working definition of life?
 - A) They must move from place to place
 - B) They all use photosynthesis
 - C) They are composed of one or more cells
 - D) They are all multicellular
2. In the scientific method, a testable, proposed explanation that follows from observations is best called a:
 - A) Theory
 - B) Hypothesis
 - C) Law
 - D) Conclusion
3. In a controlled experiment, the factor the researcher deliberately changes is usually the:
 - A) Dependent variable
 - B) Confounding variable

C) Independent variable

D) Control group only

4. The maintenance of a stable internal environment (for example, temperature or blood sugar) despite external change is called:

A) Differentiation

B) Homeostasis

C) Metabolism

D) Evolution

5. The simplest level in the hierarchy “cell → tissue → organ → organ system → organism” is the:

A) Cell

B) Organ

C) Tissue

D) Population

Module 02: Basic Chemistry

6. The **atomic number** of an element equals the number of:

A) Neutrons only

B) Electrons only in ions

C) Protons in the nucleus

D) Isotopes

7. Atoms of the same element that differ in neutron number are called:

A) Ions

B) Molecules

C) Isotopes

D) Polymers

8. A bond formed mainly by **transfer** of electrons from one atom to another, producing attraction between opposite charges, is:

- A) Covalent
- B) Hydrogen
- C) Van der Waals
- D) Ionic

9. Water is polar mainly because:

- A) It has hydrogen atoms only
- B) It cannot hydrogen-bond
- C) Shared electrons are pulled unevenly, creating partial charges
- D) It always has pH 7

10. A solution with a **lower** pH than 7 is:

- A) Basic
- B) Acidic
- C) Neutral
- D) A solution with pH above 7

Module 03: Organic Molecules

11. The element that forms the backbone of biological macromolecules is:

- A) Carbon
- B) Sodium
- C) Nitrogen
- D) Iron

12. Amino acids link by **dehydration synthesis** to form proteins, releasing:

- A) CO₂
- B) O₂
- C) ATP
- D) H₂O

13. The monomer (building block) of proteins is the:

- A) Monosaccharide
- B) Nucleotide
- C) Fatty acid
- D) Amino acid

14. Starch and glycogen are storage forms of which class of macromolecule?

- A) Lipid
- B) Carbohydrate
- C) Protein
- D) Nucleic acid

15. RNA differs from DNA in several ways; one common contrast is that RNA uses the base **uracil** instead of:

- A) Thymine
- B) Adenine
- C) Cytosine
- D) Guanine

Module 04: Cells

16. A central idea of cell theory is that **new cells** arise from:

- A) Pre-existing cells
- B) Spontaneous generation from broth
- C) Only the ER
- D) Viruses

17. Prokaryotic cells lack which structure that eukaryotic cells usually have?

- A) Ribosomes
- B) DNA
- C) A membrane-bound nucleus
- D) A plasma membrane

18. Most ATP for a typical animal cell is generated in the:

- A) Nucleus
- B) Mitochondria
- C) Lysosome
- D) Centriole

19. In plant cells but not typical animal cells, you expect to find:

- A) A cell wall outside the membrane
- B) Mitochondria
- C) Ribosomes
- D) A nucleus

20. Proteins are assembled on:

- A) Ribosomes
- B) The Golgi apparatus
- C) The smooth ER only
- D) Peroxisomes only

Module 05: Membranes

21. The basic structure of the plasma membrane is described as a:

- A) Single layer of proteins
- B) Fluid mosaic of lipids and proteins
- C) Static rigid wall
- D) DNA bilayer

22. Simple diffusion moves molecules:

- A) Only against a gradient with ATP
- B) Only in plants
- C) Only as ions
- D) With their concentration gradient, no ATP required for the diffusion step

23. In osmosis, **water** moves:

- A) Against gradients always
- B) Toward the area with lower water (higher solute) if the membrane allows water through
- C) Only into dead cells
- D) Only when sodium ions move first

24. Active transport differs from simple diffusion because it:

- A) Never uses proteins
- B) Can move substances against their concentration gradient using cellular energy
- C) Only moves water
- D) Does not depend on membrane proteins

25. Which molecules typically cross a pure lipid bilayer **least** easily without help?

- A) Small nonpolar gases
- B) Water (tiny, though polar)
- C) Large, charged ions without channels/carriers
- D) Small nonpolar molecules

Module 06: Metabolism

26. The molecule often called the cell's immediate **energy currency** is:

- A) DNA
- B) ATP
- C) Glucose only
- D) mRNA

27. Enzymes speed reactions mainly by:

- A) Lowering activation energy
- B) Raising activation energy
- C) Destroying products
- D) Removing all substrates

28. The summary outcome of **aerobic cellular respiration** (big picture) is that glucose is broken down and much energy is captured as:

- A) Only heat with no ATP
- B) ATP (and heat), with CO₂ and water among outputs
- C) Light
- D) Only lipids

29. The light-capturing reactions of photosynthesis happen in chloroplast structures associated with:

- A) Mitochondrial cristae
- B) Lysosomes
- C) Thylakoids / grana (chloroplast membranes)
- D) The nucleolus

30. Fermentation is a way some cells continue **glycolysis** when:

- A) Oxygen is abundant
- B) Photosynthesis is maximal
- C) Oxygen is limited, regenerating NAD⁺ so glycolysis can keep running
- D) DNA replicates

Module 07: Molecular Genetics

31. The phrase **central dogma** most often summarizes:

- A) DNA → RNA → protein in typical gene expression
- B) Protein → RNA → DNA
- C) Only replication
- D) Mendel's laws only

32. In DNA, adenine pairs (base-pairs) with:

- A) Guanine
- B) Cytosine
- C) Uracil
- D) Thymine

33. Transcription produces:

- A) An RNA copy of a gene, using DNA as a template
- B) A new DNA double helix only
- C) A finished protein directly
- D) Only tRNA with no mRNA

34. Translation reads which molecule to build a protein?

- A) DNA directly in the nucleus only
- B) Only rRNA with no codons
- C) mRNA at the ribosome
- D) Lipid bilayers

35. A change in the **DNA sequence** that may alter an encoded protein is a:

- A) Epigenetic mark only (same textbook definition)
- B) Mutation
- C) Gene flow event
- D) Bottleneck only

Module 08: Cellular Genetics

36. Somatic cells reproduce for growth and repair mainly by:

- A) Meiosis
- B) Fermentation
- C) Mitosis (exact copy distribution to two daughter cells)
- D) Translation

37. Meiosis produces gametes with:

- A) The same chromosome number as the parent body cell
- B) No chromosomes
- C) Only mitochondrial DNA
- D) Half the chromosome number (haploid from diploid), with genetic shuffling

38. Crossing over (exchange between homologous chromosomes) increases variation mainly during:

- A) Mitosis only
- B) Translation
- C) DNA replication only in skin
- D) Prophase I of meiosis

39. A diploid human body cell has **46** chromosomes; a normal human sperm has about:

- A) 46
- B) 23
- C) 92
- D) 0

40. The term **somatic** cell refers to:

- A) Body cells that are not gametes
- B) Only gametes
- C) Bacterial cells
- D) Dead cells only

Module 09: Inheritance Genetics

41. An organism with two identical alleles for a gene is:

- A) Heterozygous
- B) Homozygous
- C) Polyploid only
- D) Always dominant

42. In simple dominance, if **T** is tall and **t** is short, a **Tt** plant's phenotype is usually:

- A) Short
- B) Medium always
- C) Lethal
- D) Tall (dominant trait expressed)

43. A **Punnett square** is used to predict:

- A) Offspring genotypes / phenotypes from parental gametes
- B) Mitotic spindle angles only
- C) Only mutation rates
- D) Ecological carrying capacity

44. A pedigree chart tracks:

- A) Only fossils
- B) Traits through family generations
- C) Only food webs
- D) Only soil nutrient cycles

45. Mendel's idea that "heritable factors" (alleles) **segregate** into gametes underpins:

- A) Only natural selection
- B) The law of segregation (intro level)
- C) Only photosynthesis
- D) Binomial nomenclature only

Module 10: Epigenetics

46. Epigenetics in this course most closely means:

- A) Changing the DNA letter sequence permanently by CRISPR only
- B) Random gamete fusion
- C) Fermentation pathways
- D) Regulation of gene activity without changing the DNA sequence

47. DNA methylation that silences a gene region is a textbook example of:

- A) A missense mutation
- B) Mitosis
- C) An epigenetic mark (often "off")
- D) Independent assortment

48. X-inactivation in female mammals (one X compacted in each cell) helps explain, for example:

- A) Why men have Y in every liver cell only
- B) Why plants lack DNA
- C) Bacterial transformation only
- D) Calico cat patch color patterns (classic intro story)

49. Compared with a **mutation**, a typical epigenetic change to gene activity:

- A) Always changes the codon sequence
- B) Cannot respond to environment
- C) Is often reversible at the “volume knob” level of expression
- D) Only happens in viruses

50. Gene expression (big picture) refers to:

- A) Only DNA replication
- B) A gene being used to produce RNA and often protein
- C) Only photosynthesis
- D) Only drift

Module 11: Genomics & Biotechnology

51. PCR is used in teaching examples to:

- A) Amplify a targeted DNA segment in a tube
- B) Photograph mice only
- C) Replace meiosis
- D) Create ATP from sunlight

52. Gel electrophoresis separates DNA fragments mainly because:

- A) The phosphate backbone makes DNA negatively charged; fragments move through a gel toward positive
- B) Heat destroys all DNA the same way
- C) Lipids are charged like DNA
- D) Only RNA enters the gel

53. A transgenic organism carries:

- A) No DNA
- B) Only mitochondrial DNA from mother
- C) Foreign DNA introduced into its genome (GMO framing)
- D) Only fossil genes

54. CRISPR is often introduced as a tool to:

- A) Edit DNA at targeted sites (with big ethics questions)
- B) Ignore genes
- C) Make fossils
- D) Measure pH only

55. A restriction enzyme cuts DNA at:

- A) Random weather patterns
- B) Only RNA
- C) Only proteins
- D) Specific recognition sequences

Module 12: Darwin & Evolution

56. Natural selection requires heritable variation, differential reproduction, and:

- A) Goal-directed “wants” of species
- B) Lamarckian inheritance of acquired traits as the main mechanism
- C) Perfect environments only
- D) Overproduction of offspring relative to resources (broad textbook framing)

57. Descent with modification means:

- A) Each species is fixed forever
- B) Only bacteria evolve
- C) Lineages change over time; related species share ancestors with later differences
- D) Evolution cannot be tested

58. Homology (for example similar limb bone layout, different functions) supports:

- A) Identical environments only
- B) No shared history
- C) Common ancestry with adaptive change
- D) Only convergent evolution in every case

59. Fitness in evolution means roughly:

- A) Reproductive success / allele contribution to the next generation
- B) Bench press strength only
- C) Survival with zero offspring forever
- D) Size only

60. Darwin emphasized that **populations** evolve rather than individuals mainly because:

- A) Individuals rewrite DNA by thinking
- B) Fossils are not real
- C) Allele frequencies change across generations in a population
- D) Only one individual matters per species

Module 13: How Populations Evolve

61. Microevolution is defined in this course as:

- A) An individual growing taller in one year
- B) Instant new phylum in one birth
- C) Only mutation in skin cells with no offspring
- D) Change in allele frequencies in a population over time

62. The **Hardy–Weinberg** model is mainly useful as:

- A) A null / baseline to compare real allele-frequency change against
- B) Proof that evolution always speeds up
- C) A ban on mutation
- D) A species concept

63. The **founder effect** is a type of:

- A) Directional selection only
- B) Gene flow from infinite migrants always
- C) Genetic drift involving a small founding group
- D) Photosynthesis error

64. A bottleneck reduces allele diversity mainly because:

- A) Selection cannot happen
- B) Mutation stops
- C) K always equals zero
- D) A sudden severe population crash makes survivors' alleles unrepresentative by chance

65. Stabilizing selection on a trait tends to:

- A) Favor both extremes
- B) Always eliminate all variation instantly
- C) Require polyploidy
- D) Favor the middle; select against extremes

Module 14: Macroevolution

66. The biological species concept emphasizes:

- A) Reproductive isolation between groups in nature (big-picture framing)
- B) Color only
- C) Cell size only
- D) Fossil size only

67. Different firefly flash patterns that reduce mating attempts are best described as:

- A) A postzygotic barrier (hybrid infertility)
- B) Geographic separation with no mating cues
- C) Sympatric polyploidy in one generation
- D) A prezygotic behavioral barrier to reproduction

68. Mule sterility (horse $\times$ donkey hybrid) is a classic example of:

- A) Prezygotic temporal isolation between species
- B) A postzygotic barrier (reduced hybrid fertility)
- C) Convergent evolution in ungulates
- D) A prezygotic gametic barrier only

69. Allopatric speciation often begins with:

- A) Geographic separation of formerly connected populations
- B) Same habitat, no separation
- C) Only sympatry by definition
- D) Only hybrid swarms always

70. Sympatric speciation examples in intro courses often highlight:

- A) Only continental drift with zero genetics
- B) Plant polyploidy in the same region as a parent type
- C) Only fossils
- D) Only density independence

Module 15: Population & Systems Ecology

71. Early **exponential** population growth (unlimited-resource phase) is often **J**-shaped because:

- A) Growth rate is near zero always
- B) Carrying capacity is always zero
- C) Per-capita growth can stay high while numbers are still far below limits
- D) Only density-dependent rules apply on day one

72. Carrying capacity (K) in a logistic model is:

- A) The mutation rate
- B) Always infinite
- C) Gene flow only
- D) A long-run ceiling set by resources and other limits

73. A disease spreading **more easily in crowded** herds is often modeled as:

- A) Density-dependent regulation
- B) Density-independent
- C) Only abiotic freeze
- D) Carrying capacity with no biology

74. The rough ~10% **rule** between trophic levels implies:

- A) 100% of energy becomes flesh always
- B) Only a small fraction of stored energy transfers to the next consumer level (lots lost as heat/work)
- C) Decomposers violate thermodynamics
- D) Producers get energy from consumers' heat cycling perfectly

75. In ecosystems, **matter** (nutrient elements) cycles while energy from sunlight typically:

- A) Is recycled exactly like nitrogen atoms with no loss
- B) Stays only in rocks forever with zero biology
- C) Doubles at every trophic level
- D) Flows through and ends largely as heat (not recycled like chemical elements in the same way energy is framed)

Part B: Fill in the Blank (30 questions)

Write a short term or phrase in each blank (usually one or two words).

Module 01

76. In a controlled experiment, a group that does **not** receive the experimental treatment (used for comparison) is often called the

_____ group.

77. All living things must maintain organization, reproduce, respond to stimuli, undergo metabolism, and evolve as populations; many also grow and develop while maintaining

_____.

Module 02

78. A covalent bond involves **sharing** pairs of _____ between atoms.

79. A neutral atom has equal numbers of protons and _____ (in the standard model for this course).

Module 03

80. The chemical bond that links amino acids in a chain is a _____ bond.

81. Fats and oils belong to the macromolecule class called _____.

Module 04

82. Membrane-bound sacs of digestive enzymes that break down macromolecules in many eukaryotic cells are _____.

83. The control center that holds most DNA in a eukaryotic cell is the _____.

Module 05

84. Transport proteins that open the door for molecules **without** using ATP for that step (along a gradient) perform _____ transport.

85. The phospholipid tails of the bilayer are _____
(hydrophilic or hydrophobic)—choose the correct term.

Module 06

86. The anaerobic portion of glucose breakdown that happens in the cytoplasm (and yields a little ATP and pyruvate) is

87. The pigment that captures light energy in plants and many photosynthetic organisms is

Module 07

88. During DNA replication, the enzyme class that **matches** incoming nucleotides to the template strand (5'→3' synthesis on a daughter strand) is

polymerase (family name only is fine).

89. A **three**-nucleotide mRNA code that specifies an amino acid (or stop) is a

Module 08

90. The stage of mitosis where **sister chromatids** are pulled apart toward opposite poles is

91. Homologous chromosomes pair up during

I of meiosis.

Module 09

92. An observable trait (tall, short, etc.) is called the

93. When both alleles are visibly expressed in the heterozygote (classic AB blood-type framing), inheritance is often called

Module 10

94. Methyl groups added to DNA can often **silence** nearby genes as an

mark (adjective form used in class).

95. Proteins that help **turn transcription on or off** near promoters are often called transcription

Module 11

96. A **plasmid** is a small, circular **extra** piece of DNA often used in bacterial _____ work (one-word lab context is fine).

97. To cut and paste a gene, labs often use **restriction** enzymes and

enzyme to seal DNA back together.

Module 12

98. Heritable differences among individuals in a population are called

99. When unrelated lineages evolve **similar** traits because of similar selection pressures (not shared recent ancestry), it is

evolution.

Module 13

100. When chance events change allele frequencies (especially in small populations), the umbrella term is genetic

101. If **two** alleles A and a have frequencies **p** and **q** in a diploid population at Hardy–Weinberg equilibrium, the expected heterozygote frequency is _____ (symbolic answer).

Module 14

102. Barriers that block mating or fertilization **before** a zygote forms are _____ barriers.

103. A **reduced hybrid** survival or fertility after a zygote forms is a _____ zygotic barrier category.

Module 15

104. All the organisms of one species living in an area at one time form a(n) _____

105. Organisms that make organic matter from inorganic carbon (often via photosynthesis) are **primary** _____ (trophic role).

Part C: Free Response (30 questions)

Answer in complete sentences. Each item is tagged with its home module.

106. (Module 01) State two characteristics of life and **briefly** explain each in your own words.

107. (Module 01) What is the difference between a **scientific theory** and a **hypothesis** in everyday lab language?

108. (Module 02) Explain how water's polarity helps it dissolve **ionic** compounds such as table salt at the molecular level (big picture, no deep math).

109. (Module 02) Why does a **buffer** matter in blood and cellular chemistry (one or two sentences of "what it does")?

110. (Module 03) Name the **four** major classes of biological macromolecules and, for **one** class you choose, name its monomer type.

111. (Module 03) What happens to a protein's function when it is **denatured**, and what is one common cause of denaturation?

112. (Module 04) Compare **prokaryotic** and **eukaryotic** cells in **two** sentences (nucleus + relative complexity).

113. (Module 04) What is the main job of the **mitochondria**, and where in the cell does it do that work?

114. (Module 05) Contrast **passive** and **active** transport across a membrane with **one example** of each.

115. (Module 05) In plain language, what happens to a cell placed in a **very salty** external solution if the membrane is permeable to water only (animal cell framing)?

116. (Module 06) Write one sentence linking **photosynthesis** and **cellular respiration** as complementary carbon/energy themes on Earth.

117. (Module 06) Why do enzymes stop speeding a reaction if the temperature becomes **too high**?

118. (Module 07) Describe **transcription** and **translation** in **two** short sentences, using the words **DNA**, **RNA**, **ribosome**, and **protein** sensibly.

119. (Module 07) Why is **DNA replication** described as **semiconservative** at the big-picture level?

120. (Module 08) Contrast the **goal** of mitosis versus meiosis in a multicellular animal (what each type produces).

121. (Module 08) Name **two** ways meiosis increases **genetic variation** besides random fertilization.

122. (Module 09) In a monohybrid cross $Aa \times Aa$ with complete dominance, what are the expected **phenotype** ratios in offspring?

123. (Module 09) Define **heterozygous** and give a **letter example** (any gene).

124. (Module 10) In one short paragraph, explain how **epigenetics** differs from a **mutation**.

125. (Module 10) Give one **realistic** example of how environment **without** changing DNA letters might still influence **trait expression** at the gene-regulation level.

126. (Module 11) In two sentences, what is a **GMO** in food/ag contexts, and why do people disagree ethically?

127. (Module 11) What does **gel electrophoresis** show a biologist about DNA samples (smaller vs larger fragments)?

128. (Module 12) Explain **natural selection** in four short sentences aimed at a friend who has never taken biology.

129. (Module 12) Name **two** independent lines of **evidence** that life shares common ancestry and **briefly** say what each shows.

130. (Module 13) Define **microevolution** and give **one** example of a mechanism that can change allele frequencies **other** than natural selection.

131. (Module 13) Contrast **directional**, **stabilizing**, and **disruptive** selection in one sentence **each**.

132. (Module 14) Contrast **allopatric** and **sympatric** speciation with **one sentence each**.

133. (Module 14) Give **one** example of a **prezygotic** barrier and **one** of a **postzygotic** barrier.

134. (Module 15) Contrast **exponential** vs **logistic** population growth; mention **carrying capacity** in your answer.

135. (Module 15) Explain the **10% energy rule** and **why** long food chains support fewer top predators than short chains from the same plant base.

End of Practice Test 05